Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21321
F-421
B.C.A. (Fifth Semester)
Examination, March-2022
(New Course)
Paper - BCA-501-N
Computer Graphics And Animation

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are various elements of computer graphics?

   (b) Discuss different image formats.

2. (a) How raster scan is different from random scan?

   (b) Describe different display technologies.

P.T.O.

https://www.mjpruonline.com

3. (a) Explain any line drawing algorithm.

   (b) What is antialiasing?

4. (a) Explain various picture construction techniques.

   (b) Write in brief about the following.

   (i) Clipping

   (ii) 3–D transformation

5. (a) How to draw circle and ellipse? Explain with proper example.

   (b) Differentiate 2–D transformation from 3–D transformation.

6. (a) How Animation is implemented in computer graphics?

   (b) Elaborate the 3–D viewing algorithm.

7. (a) What do you mean by the term "MORPHING"? Write down its practical application giving one suitable example.

https://www.mjpruonline.com

(b) Find the new coordinate of triangle A(0, 0), B(1, 1), C(5, 2) after it has been magnified to twice of its size.

8. (a) Define translation, rotation and scaling.

(b) Define Polygon clipping.

(c) Define recursively defined B–Splines.

(d) Define scalars, point and vectors.

https://www.mjpruonline.com